

Advanced Software Engineering

Activity 6 – Factory Pattern – Spring 2024

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Assume that you work for a company that is creating the next generation graphical user interfaces on top of the Java Swing Framework. Some of the interns in the company came up with a rough working prototype of the system. Your manager has asked you to refactor their code to achieve the following:

F1. Currently only Windows, Labels, and TextBoxes are supported in the framework. Your manager would like to see the system support additional GUI widgets such as Button, List, and so on, but at a low maintenance cost. As a proof of concept, having a support for a regular **command button** is just fine for this feature. Implementation of the click functionality is not critical at the moment.

F2. Currently the system supports Windows and Ubuntu operating systems. Your manager would like to support more operating systems in the future, but again, with low maintenance hassle. As a proof of concept, having a support for the Mac OS would be just fine for this feature.

Your manager would like to have an end product that requires minimum/no changes to the core logic of the system after refactoring.

Implement your solution and describe what have been done in details.